Data Extractor A

converts input datasets into a set of actionable tables

INPUT

papers_1990-2024.csv

```
year,title,track,nAuth
1990,Volume rend…,Vis,3
2024,LLM4Vis…,VIS,11
```

citations.json

```
{"1990": [12, 4, …], …}
```

ACTIONABLE TABLES

year	track	papers
1990	Vis	53
2007	InfoVis	104
2024	VIS	174

3,620 rows · every field kept

+ derived analysis tables

Persona Explorer B

designs questions to facilitate exploration based on two personas

PERSONAS

domain expert
depth, mechanism, exceptions

curious reader
orientation, scale, comparison

QUESTIONS

guided — main narrative

- How did paper volume change?
- Which tracks grew fastest?

open — reader explores

- Which topics co-occur?
- Who collaborates most?

distinct questions per persona

Visualization Generator C

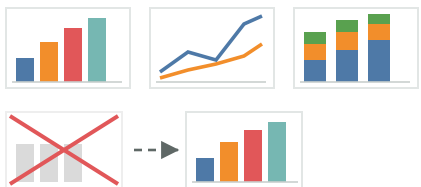
creates and frames a set of panels to address the designed questions

PANEL SPECIFICATIONS

```
{"mark":"bar",
 "x": {"field":"year"},
 "y": {"aggregate":"count"},
 "transform": [{"filter":…}],
 "summary":"3× since 1990"}
```

all panels + framing, one call

FRAMED PANELS



rejected: filter matched 0 rows
redrawn from the stated reason

Report Designer D

lays the panels out into a narrative structure

NARRATIVE LAYOUT

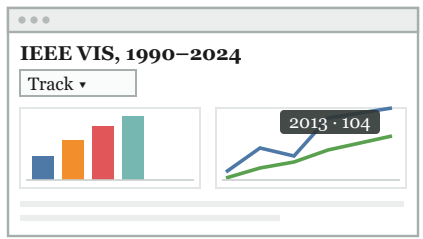
guided opening

- panel 1 · volume over time
- panel 2 · growth by track

open exploration

- panel 3
- panel 4

REPORT



Validator *runs at every stage*

Stage A · Data Extractor
- Preserves source data
- Verifies analysis coverage

Stage B · Persona Explorer
- Ensures question coverage
- Validates data grounding

Stage C · Visualization Generator
- Rejects invalid charts
- Repairs visual problems

Stage D · Report Designer
- Tests rendered output
- Verifies interactions

a failure at any stage returns to the model as a repair instruction, and every stage keeps a valid result to fall back to